Isogeny-class trio: three 61-bit roots for a whole-class experiment

Date: 2026-10-17. Protocol: `PROTOCOL.md`, frozen before the B and P searches; its 2026-10-17 amendment records the user's choice of the 61-bit trio as the primary picks. PDF: `REPORT.pdf`. Data: `data/`. Tool: `selector/` (native Rust, no dependencies) and `identity/` (the repository's native ICV1 producer).

Question

The user asked for three curves, one binary, one Koblitz and one prime-field, between 41 and about 66 bits (preferably above 45), with shapes that emulate standardized curves. Each should have Frobenius conductor `f > 1`, so that its isogeny class has volcanoes with more than one level, as the root of an experiment that iterates over every curve in the isogeny class. After seeing both candidate trios the user chose the 61-bit one.

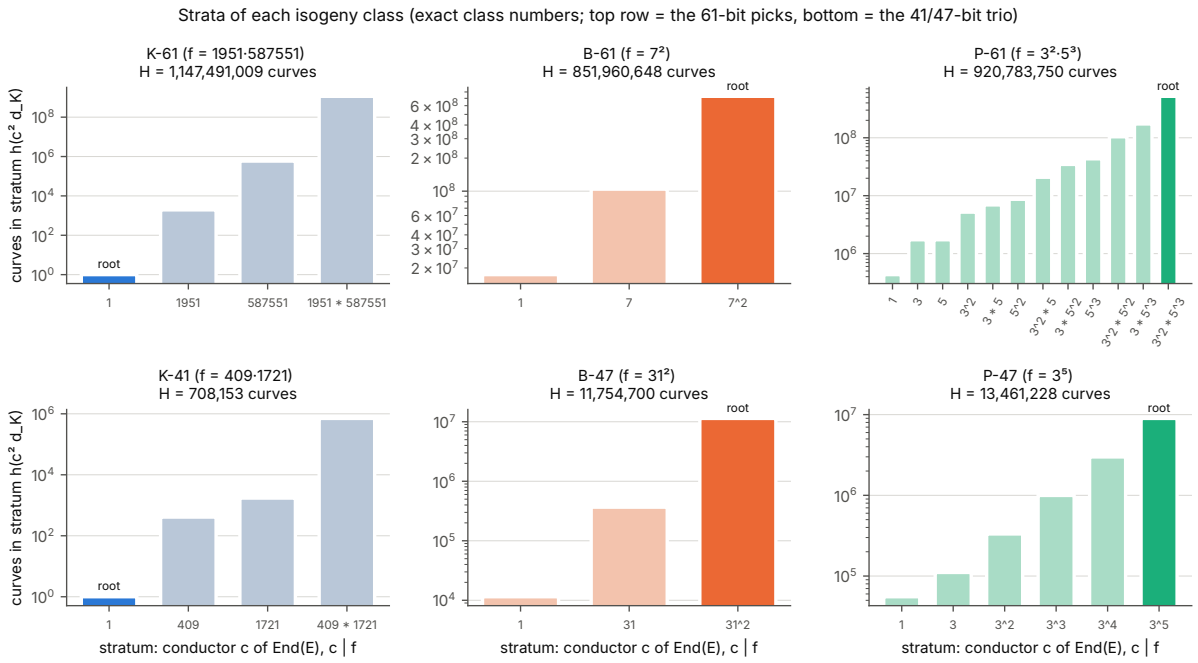
Picks (61-bit trio, chosen by the user)

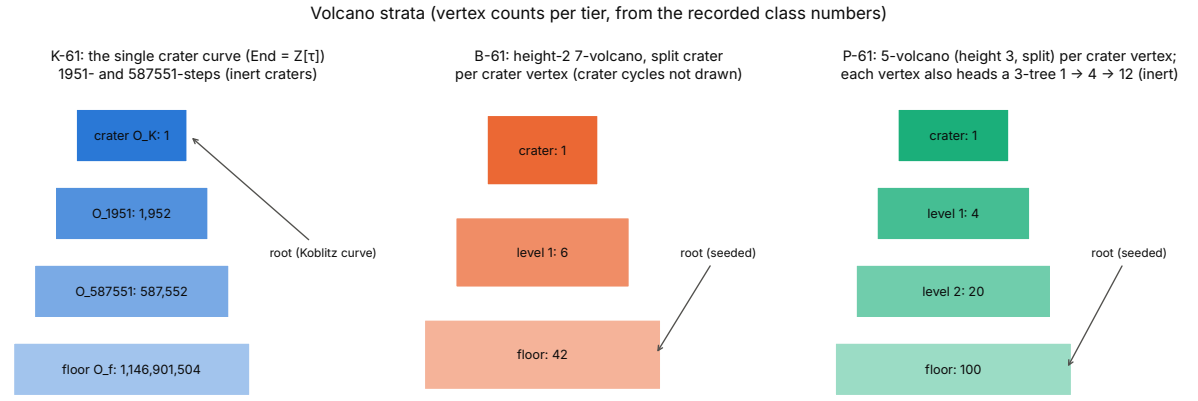
	Koblitz (K-shape)	Binary (B-shape)	Prime (P-shape)
ICV1 slug	`icv1-f2m61-t158598901-ab42b6c5`	`icv1-f2m61-tm2657393969-b9ff0fb0`	`icv1-fp61-tm813574577-a3b71255`
Emulates	K-233/283/409/571 (`a = 0`); cofactor is not standard (none exists, finding 1)	B-163…B-571 (`a = 1`, seeded `b`, cofactor 2)	P-521 (`p = 2^k − 1` Mersenne, `a = −3`, seeded `b`, prime order)
Field	`GF(2^61)`, `x^61 + x^5 + x^2 + x + 1`	same	`GF(p)`, `p = 2^61 − 1`
Model	`y^2 + xy = x^3 + 1`	`y^2 + xy = x^3 + x^2 + b`, `b = 0x9c60f836d06fc8`	`y^2 = x^3 − 3x + b`, `b = 0x54aa66b458b77ec`
Seed	— (fixed by the family)	`isogeny-class-trio/2026-10-17/B-61/b/12079`	`isogeny-class-trio/2026-10-17/P-61/b/11526`
`#E`	`4 · 3539 · 162888033982417`	`2 · 1152921505935543961`	`2305843010027268529` (prime)
Prime subgroup	48 bits (cofactor 14156)	61 bits	62 bits
Trace `t`	158598901	−2657393969	−813574577
`d_K`	−7	−900303761090047	−6764617042579
Conductor `f`	`1951 · 587551`	`7^2`	`3^2 · 5^3`
Volcanoes	`l = 1951`, `l = 587551`: height 1, inert	`l = 7`: height 2, split crater	`l = 3`: height 2, inert; `l = 5`: height 3, split
Strata (`c | f`)	4	3	12
Class size `H`	**1,147,491,009**	**851,960,648**	**920,783,750**
Root's position	crater (`End = Z[τ] = O_K`)	floor (`End = Z[π]`, conductor 49)	floor of both volcanoes (`End = Z[π]`, conductor 1125)
Generator	`(0x5cc1d6ba365013e, 0x171d9a59a2d31eb9)` (registry, `EC1N61Ce0hd8c15448081e`)	`(0xc17a1d9ae3996ee, 0x9bcf1bc6ebcb05)`	`(0x112c917afe3d4fd3, 0xef721929b08bfaa)`

Stratum sizes (curves with `End = O_c`, exact):

- K-61: `c = 1`: 1; `1951`: 1,952; `587551`: 587,552; `1951·587551`: 1,146,901,504.
- B-61: `c = 1`: 17,386,952; `7`: 104,321,712; `7^2`: 730,251,984.
- P-61: `c = 1`: 433,310; `3`: 1,733,240; `5`: 1,733,240; `9`: 5,199,720; `15`: 6,932,960; `25`: 8,666,200; `45`: 20,798,880; `75`: 34,664,800; `125`: 43,331,000; `225`: 103,994,400; `375`: 173,324,000; `1125`: 519,972,000.

K-61 caveat for the whole-class pass. The Koblitz curve is the only curve of its class on the crater, and every small-`l` isogeny from it is a loop (finding 2). The `O_1951` stratum (1,952 curves) is one degree-1951 step away, with kernel x-coordinates in `F_{q^975}`. The other 1,147,489,056 curves (99.9998% of the class) need a degree-587551 step whose kernel x-coordinates lie in `F_{q^58755}`, a field of about 3.6 million bits. That step is the open obstruction for iterating over this class; B-61 and P-61 have no such step (degrees 7, 3 and 5).





41/47-bit trio (the first proposal, kept for reference)

	K-41	B-47	P-47
ICV1 slug	`icv1-f2m41-tm2308219-7f48b14a`	`icv1-f2m47-t4276051-826c5755`	`icv1-fp47-tm8785349-43e36ce5`
Field	`x^41 + x^3 + 1`	`x^47 + x^5 + 1`	`p = 2^47 − 115`
Model	`a = 0, b = 1` (cofactor 4, the only standard-cofactor K in range)	`a = 1, b = 0x52e3db551eb3` (seed index 28948)	`a = −3, b = 0xda8a554f2ba` (seed index 117828)
Subgroup	40 bits	46 bits	48 bits
`f`	`409 · 1721`	`31^2` (inert crater)	`3^5` (split crater)
Strata, `H`	4, 708,153	3, 11,754,700	6, 13,461,228
Root	crater	floor	floor

Their stratum sizes: K-41 1 / 410 / 1,722 / 706,020; B-47 11,468 / 366,976 / 11,376,256; P-47 55,396 / 110,792 / 332,376 / 997,128 / 2,991,384 / 8,974,152.

Prime-conductor-gap set (61 bits, class-number-one CM)

The user added a criterion: a **large prime conductor gap**, the Frobenius conductor being a single large prime `f = l`, so the class is a crater (`End = O_K`) and a floor (`End = Z[π]`) at index `l`. Seeded random coefficients essentially never give a large `l` (the largest conductor prime among 1,526 seeded 61-bit prime-subgroup curves was 109), so the user approved a CM construction (`PROTOCOL-gap.md`, frozen before it ran). Every curve below has prime order, sits on the crater, and has exactly two strata. The order is certified by `n·P = O` with `n` prime and `n > 4√p`; `src/bin/isogeny_walk.rs` reproduces each trace, `Δ = d·l^2` and the prime order (`data/walker-class-cm-gap-d*.txt`).

	`d`	Form	`p`	Gap `l`	Floor `h(d l^2)`	Class size
★ `icv1-fp61-t205-983ecb51`	−3	`y^2 = x^3 + 5` (secp256k1 form, `j = 0`)	`0x1ffffff43f9a1739`	1,733,413,037	584,471,012	584,471,013
★ `icv1-fp61-t7175-7e38f752`	−19	`y^2 = x^3 − 3x + 0x14eb628f40fdf1f3` (P-curve form)	`2305842959800706971`	696,735,691	696,735,692	696,735,693
`icv1-fp61-t727-ddf032b6`	−43	P-curve form	`2305842677174433443`	463,138,201	463,138,200	463,138,201
`icv1-fp61-t2279-df6b5e41`	−67	P-curve form	`2305842976999134887`	371,028,661	371,028,660	371,028,661
`icv1-fp61-t95-4d1bb712`	−163	P-curve form	`2305842838172855237`	237,876,229	237,876,228	237,876,229

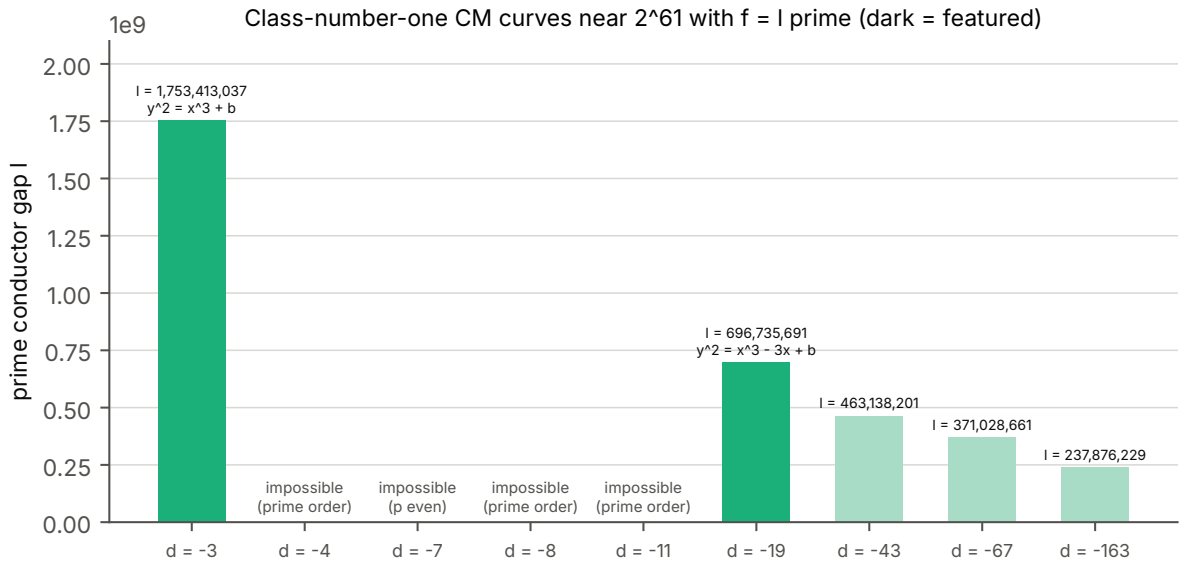
Full parameters, generators and strata are in `data/cm-gap.jsonl`.

Impossible discriminants (derived, recorded in the protocol notes):

- `d = −4, −8`: `t` must be even, so `n = p + 1 − t` is even and prime order is impossible.
- `d = −7`: odd `t` and odd `l` make `t^2 + 7 l^2 ≡ 0 (mod 8)`, so `p` would be even.
- `d = −11`: `p ≡ t^2 + 2 l^2 (mod 3)`, so `p` prime forces `3 | t` and then `3 | n`.

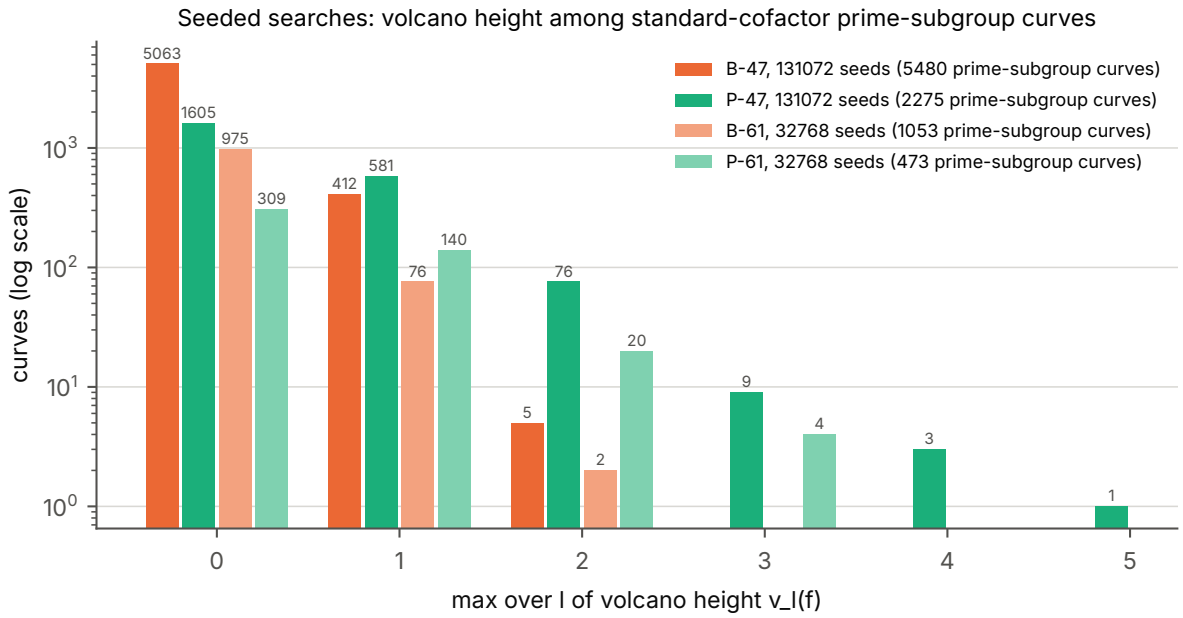
Binary member: the Koblitz pick. Over `GF(2^m)` with `m` prime, `(π) = 𝔭^m`. So either `𝔭` is principal, which gives `8 = a^2 + |d| b^2`, `d = −7` and a Koblitz class; or `m | h(d_K)`. Class-number-one CM therefore gives nothing new in characteristic 2, and the binary large-gap curve is `icv1-f2m61-t158598901-ab42b6c5` (prime gaps 1951 and 587551). A binary CM curve outside the Koblitz classes needs a Hilbert class polynomial of degree at least 61, which this study does not build.

What a whole-class pass needs here. The vertical step has degree `l` (about `2^28` to `2^31`), and its kernel x-coordinates lie in extensions of degree about `l/2` to `l/8` (`kernel_x_field_degree` in the data). The crater curve's small-`l` isogeny graph is a single vertex with loops. As with K-61, the floor (essentially the whole class) is not reachable from the crater by any isogeny this repository can compute. It has to be constructed directly (a ring class polynomial of degree about `l`) or reached by a new large-degree isogeny method. This is the price of a large prime gap, and the experiment design has to plan around it.



Findings

1. **No prime-degree Koblitz curve has the standard cofactor between 2^42 and 2^66** (derived, exhaustive). The prime degrees in range are 41, 43, 47, 53, 59 and 61; of the 12 curves only `K_0` over `GF(2^41)` has `#E = 4r` with `r` prime (`data/koblitz.jsonl`, figure). Above 45 bits the largest prime-degree subgroup is `K_0` over `GF(2^61)` (cofactor 14156, 48-bit subgroup), the K pick.
2. **A prime-degree Koblitz class has height-1 volcanoes at large `l` only** (derived; classical). `f = |U_m|`, the Lucas sequence of `τ^2 − μτ + 2`. For prime `m ≠ 7` every prime `l | U_m` has rank of apparition `m`, so `l ≡ ±1 (mod m)` and `l ≥ 2m − 1`; the table bears this out (1951, 587551 ≡ −1, −1 mod 61; 409, 1721 ≡ −1, −1 mod 41). The Koblitz curve is the only curve in its class with `End = O_K` (`h(−7) = 1`), so for every small `l` its isogeny graph is one vertex with loops: **the rest of the class is reachable only through the vertical steps of degree `l | f`.** For K-61 those are degree 1951 and 587551, with kernel x-coordinates in `F_{q^975}` and `F_{q^58755}` (computed: the order of `±t/2` in `F_l^*/±1`).
3. **Binary curves over odd-degree fields have conductor primes `l ≡ ±1 (mod 8)` only** (derived; confirmed by the searches). `l | f` needs `t^2 ≡ 2^(m+2) (mod l)`, i.e. 2 a square mod `l`. So 3- and 5-volcanoes never occur for B- or K-shape curves; the smallest possible vertical degree is 7. Every conductor prime in the B-61 and B-47 searches was in {7, 17, 23, 31, 41, 47, 71, 73, 79, …} (`data/tally-B61.json`, `data/tally-B47.json`). For the same reason only 2 of 1,053 standard-cofactor B-61 curves have a height-2 volcano, against 24 qualifying of 473 for P-61.
4. **Both seeded roots sit on the floor** (proved by the tests below), as most curves do: the floor holds 85.7% of B-61's class and 56.5% of P-61's. Their crater representatives are 2 (B-61) and 2 + 3 (P-61) ascending isogenies away.



Method

- **Koblitz.** `#E(GF(2^m)) = 2^m + 1 − V_m` and `f = |U_m|` from the Lucas sequences of τ. The selector asserts `t^2 − 2^(m+2) = −7 U_m^2` for every row. The orders of K-41 and K-61 match their registry entries (`icv1-f2m41-tm2308219-7f48b14a`, `icv1-f2m61-t158598901-ab42b6c5`), and the native ICV1 producer reproduces their slugs.
- **B and P searches.** Coefficients come from SHA-256 of public seed strings, as in X9.62/FIPS 186 seeding. The selector counts points by BSGS over the Hasse window on seeded points, using `#E ≡ 2 (mod 4)` for B. A curve is kept only when the match is unique and `r = #E/h` is prime. `r > 4√q` then makes `#E` the only multiple of `r` in the window, so each kept order is proved. Not one count was ambiguous across the 327,680 curves.
- **Selection rule (frozen).** `f > 1`, `f` 61-smooth, a volcano of height ≥ 2. Rank by height, then number of strata, then lowest seed index.
- **Class invariants.** `Δ = t^2 − 4q` is factored to `f^2 d_K`. `h(d_K)` is computed exactly by counting reduced forms (`O(√|d_K|)`; cross-checked against brute force for every fundamental `|D| < 4000`). Each stratum's `h(c^2 d_K)` comes from the conductor formula, and `H = Σ_{c|f} h(c^2 d_K)` (Schoof 1987, Thm 4.6).
- **Root levels.**
  - **K:** `τ ∈ End(E)` gives `End = O_K` (crater).
  - **B:** for odd `l | f`, above the floor π acts on `E[l]` as the scalar `t/2`, so all `(l^2 − 1)/2` x-coordinates of `E[l]` lie in `F_{q^k}`, with `k` the order of `±t/2` in `F_l^*/±1`. On the floor `π − t/2` is a non-zero nilpotent, so only the eigenline's `(l − 1)/2` x-coordinates do. The test is therefore `deg gcd(ψ_l, x^(q^k) − x)` using characteristic-2 division polynomials. B-47 gives 15 of 480 (floor). Control: over `GF(2^7)` the Koblitz class has `f = 7`, and the test flags exactly the Koblitz curve as above the floor and the other 7 class members as floor (`floor_test_control_koblitz_m7`).
  - **P:** a 3-isogeny BFS (roots of `ψ_3`, Vélu) finds that the root has exactly one rational 3-isogeny. The repository walker independently agrees: `src/bin/isogeny_walk.rs`, with the same trace, `Δ`, a 3-volcano of depth 5, and one kernel-certified 3-edge from the root (`data/walker-class-P47.txt`). For P-61 the walker records one 3-edge and one 5-edge from the root, so the root is on both floors.
- **Tests.** `cargo test --release` in `selector/` runs 15 tests:

- BSGS against naive counts, in both characteristics;
- characteristic-2 3-isogenies against brute-force orders, plus their duals;
- division polynomials against rational torsion;
- class numbers against brute force;
- SHA-256 vectors;
- the floor-test control.

Status of each statement

Statement	Status
Orders, traces, `f`, `d_K` of all six curves	computed (native); order proved by `r > 4√q` (B, P) or exact formula (K)
`h(d_K)` and stratum sizes, `H`	computed exactly (native form count); formula classical
Root levels	B, P: proved by the tests above; P cross-checked by the walker; K: theorem
No standard-cofactor prime-degree Koblitz in 42–66	exhaustive enumeration
`l ≡ ±1 (mod m)` for Koblitz, `±1 (mod 8)` for binary	classical arguments, written above; consistent with all data
Seeded picks are optimal under the rule	within the frozen seed ranges only

No theorem in this report is new, so there is no Lean obligation (AGENTS.md, "Mathematical results should include Lean proofs"). Every structural fact used is classical and its argument is written out above.

What the whole-class experiment will need (not done here)

- **P-61** (921 M curves, 12 strata): the repository walker supports `p = 2^61 − 1` and every odd `l`, and the vertical steps have degree 3 and 5. It holds about 87 KB per curve, so a full pass needs a compact class-group enumeration such as the `p61_class` segment stream (`research/p61_isogeny_class_20261014/`, same field, another class).
- **B-61** (852 M curves): there is no binary isogeny walker (`docs/isogeny-walk/README.md`: "Prime fields and odd l only").
  - It needs characteristic-2 `l`-isogenies: degree 7 for the vertical steps (kernel x-coordinates in `F_{q^3}`), and split primes (`(d_K/l) = 1`) for horizontal steps.
  - This selector already has `ψ_l` and 3-isogeny Vélu in characteristic 2.
  - 2-isogenies are horizontal here (2 splits, since `d_K ≡ 1 mod 8`).
- **K-61** (1.15 G curves): a degree-1951 step (kernel field `F_{q^975}`) reaches 1,952 curves. The remaining 1,147,489,056 need a degree-587551 step with kernel field `F_{q^58755}`, which no tool here can compute yet. That is the open obstruction for this class.

Reproduce

```
cd research/isogeny_class_trio_20261017/selector
cargo test --release && cargo build --release
B=target/release/isogeny-class-trio-selector
$B koblitz --min 41 --max 64                       > ../data/koblitz.jsonl
$B binary --m 47 --count 131072 --threads 4        > ../data/search-B47.jsonl 2> ../
data/search-B47.summary.json
$B prime --bits 47 --count 131072 --threads 4      > ../data/search-P47.jsonl 2> ../
data/search-P47.summary.json
```

```
$B binary --m 61 --count 32768 --threads 4         > ../data/search-B61.jsonl 2> ../
data/search-B61.summary.json
$B prime --bits 61 --count 32768 --threads 4       > ../data/search-P61.jsonl 2> ../
data/search-P61.summary.json
$B analyze-binary --m 61 --a2 0 --b 1 --koblitz    > ../data/pick-K61.json
$B analyze-binary --m 61 --a2 1 --b 0x9c60f836d06fc8 --seed isogeny-class-
trio/2026-10-17/B-61/b/12079 > ../data/pick-B61.json
$B analyze-prime --p 2305843009213693951 --b 0x54aa66b458b77ec --seed isogeny-class-
trio/2026-10-17/P-61/b/11526 > ../data/pick-P61.json
$B check-point --m 61 --a2 0 --b 1 --r 162888033982417 --gx 0x5cc1d6ba365013e --gy
0x171d9a59a2d31eb9
$B tally --in ../data/search-B61.jsonl             > ../data/tally-B61.json
$B cm-gap                                         > ../data/cm-gap.jsonl
# (41/47-bit trio: same commands with --m 41 / 47, --p 140737488355213 and the seeds
in the table)
```

Wall time on the 4-core session host: B-47 search 1 m 59 s, P-47 search 4 m 18 s, B-61 and P-61 searches about 27 min together (sharing the host with a build), analyses under 5 s each. Figures: `python3 figures/plot.py` (plots recorded values only). The curves are registered in `docs/curves/registry.json` from `curves.json`.